H524 Introduction to Biostatistics

Week 4: Confidence Intervals

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Outline

Quick Review: Sampling Distributions

Last week we covered:

- Binomial and normal distributions
- Standardization and Z-scores
- Central Limit Theorem: $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ for large n
- Standard error: $\text{SE} = \frac{\sigma}{\sqrt{n}}$
- Introduction to confidence intervals

This Week: Deep dive into confidence intervals (Chapter 9)

Reading: Pagano & Gauvreau Chapter 9

The Problem of Point Estimation

Scenario: We measure cholesterol in 25 randomly selected adults

- Sample mean: $\bar{x} = 205$ mg/dL
- Sample SD: $s = 42$ mg/dL

Question: What is the true population mean μ?

Point estimate: Our best guess is $\bar{x} = 205$ mg/dL

But...

- Different samples would give different $\bar{x}$ values
- We know there's sampling variability
- How confident are we in this estimate?
- What's the range of plausible values for μ?

Solution: Use a *confidence interval* instead of just a point estimate

What is a Confidence Interval?

Definition: A range of values that likely contains the unknown population parameter

General Form:

$$\text{Estimate} \pm \text{Margin of Error}$$

$$\text{Estimate} \pm (\text{Critical Value}) \times (\text{Standard Error})$$

For a population mean:

$$\bar{x} \pm t_{\alpha/2,n-1} \times \frac{s}{\sqrt{n}}$$

Components:

- $\bar{x}$ = sample mean (point estimate)
- s = sample standard deviation
- n = sample size
- $t_{\alpha/2,n-1}$ = critical value from t-distribution

Confidence Level and Interpretation

Confidence level: Usually 95%, but can be 90%, 99%, etc.

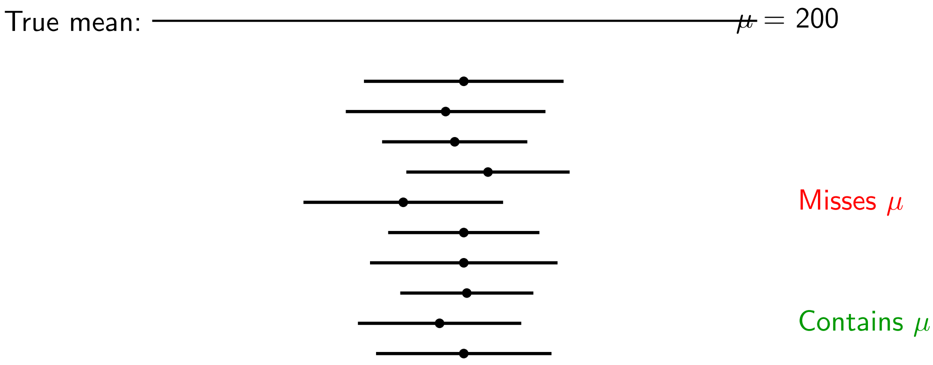
Correct interpretation of a 95% CI:

- "We are 95% confident that the true population mean is between [lower, upper]"
- If we repeated this procedure many times, 95% of the intervals would contain μ
- The method is reliable 95% of the time

What 95% confidence does NOT mean:

- $\times$ "There's a 95% probability μ is in this interval"
  - μ is fixed (not random); it's either in or out
- $\times$ "95% of the data falls in this interval"
  - CI is about the mean, not individual values
- $\times$ "There's a 95% chance the next sample mean will be in this interval"
  - Each new sample gets its own CI

Visualizing Confidence Intervals



If we took 20 samples:

- Each sample gives a different CI
- About 95% of CIs (19 out of 20) contain μ
- About 5% (1 out of 20) miss μ

When to Use the t-Distribution

Recall from Week 3:

- If σ is known: use Z-distribution (rare in practice)
- If σ is unknown: use t-distribution (almost always)

The t-distribution:

- Bell-shaped like normal, but heavier tails
- Accounts for uncertainty in estimating σ with s
- Determined by degrees of freedom: $\text{df} = n - 1$
- As n increases, t approaches normal

Assumptions for t-interval:

1. Random sample from the population
2. Population is approximately normal, OR n is large ($n \geq 30$)
3. Observations are independent

Robust to mild departures from normality, especially for larger n

Computing a t-Based Confidence Interval

Step 1: Calculate sample statistics

- Sample mean: $\bar{x}$
- Sample SD: s
- Sample size: n

Step 2: Compute standard error

$$\text{SE} = \frac{s}{\sqrt{n}}$$

Step 3: Find critical value

- Degrees of freedom: $\text{df} = n - 1$
- For 95% CI: $t_{0.975,\text{df}}$ (leaves 2.5% in each tail)
- Use t-table or R: `qt(0.975, df)`

Step 4: Calculate margin of error

$$\text{ME} = t_{\alpha/2,\text{df}} \times \text{SE}$$

Example: Blood Pressure Study

Research question: What is the mean systolic blood pressure for adults with hypertension?

Data: Random sample of $n = 25$ hypertensive adults

- Sample mean: $\bar{x} = 128.4$ mmHg
- Sample SD: $s = 12.6$ mmHg

Calculate 95% CI:

Step 1: Standard error

$$\text{SE} = \frac{12.6}{\sqrt{25}} = \frac{12.6}{5} = 2.52 \text{ mmHg}$$

Step 2: Critical value (df = 24)

$$t_{0.975,24} = 2.064$$

Step 3: Margin of error

$$\text{ME} = 2.064 \times 2.52 = 5.20 \text{ mmHg}$$

Step 4: Confidence interval

Interpreting the Blood Pressure CI

Result: 95% CI = (123.2, 133.6) mmHg

Correct interpretation:

- "We are 95% confident that the true mean systolic blood pressure for adults with hypertension is between 123.2 and 133.6 mmHg"
- Based on this sample, plausible values for μ range from 123.2 to 133.6
- If we repeated this study many times, 95% of our CIs would contain μ

Clinical context:

- Normal BP: $<$ 120 mmHg systolic
- Elevated: 120-129 mmHg
- Hypertension Stage 1: 130-139 mmHg
- The entire CI is above normal range
- This provides evidence that the population mean is elevated

Example: Drug Efficacy Trial

Scenario: A new pain medication is tested on 16 patients

Outcome: Pain reduction score (0-100 scale)

- Sample mean: $\bar{x} = 45.2$ points
- Sample SD: $s = 8.4$ points
- Sample size: $n = 16$

Calculate 95% CI:
Standard error: $\text{SE} = \frac{8.4}{\sqrt{16}} = \frac{8.4}{4} = 2.10$
Critical value: $t_{0.975,15} = 2.131$
Margin of error: $\text{ME} = 2.131 \times 2.10 = 4.48$
95% CI: $(45.2 - 4.48, 45.2 + 4.48) = (40.7, 49.7)$ **points**

Interpretation: We are 95% confident that the true mean pain reduction is between 40.7 and 49.7 points.

What Makes a CI Wider or Narrower?

Recall: $\text{CI} = \bar{x} \pm t_{\alpha/2,n-1} \times \frac{s}{\sqrt{n}}$

Three main factors affect width:

1. **Sample size (n)**
   - Larger $n \Rightarrow$ smaller SE $\Rightarrow$ narrower CI
   - Width decreases by factor of $1/\sqrt{n}$
   - To halve width, need $4\times$ the sample size
2. **Sample variability (s)**
   - Larger $s \Rightarrow$ larger SE $\Rightarrow$ wider CI
   - More variable data $=$ more uncertainty
   - Cannot control s (it's a property of the population)
3. **Confidence level ($1 - \alpha$)**
   - Higher confidence $\Rightarrow$ larger $t \Rightarrow$ wider CI
   - Trade-off: confidence vs. precision

Effect of Sample Size

Same data, different sample sizes:

Suppose $\bar{x} = 100$, $s = 20$ across all samples

n	SE	df	$t_{0.975}$	ME	95% CI
10	6.32	9	2.262	14.3	(85.7, 114.3)
25	4.00	24	2.064	8.3	(91.7, 108.3)
50	2.83	49	2.010	5.7	(94.3, 105.7)
100	2.00	99	1.984	4.0	(96.0, 104.0)

Observations:

- As n increases, SE decreases
- As n increases, t approaches 1.96 (Z-value)
- CI width decreases substantially with larger n
- Diminishing returns: 10→25 gives bigger gain than 50→100

Effect of Confidence Level

Same data, different confidence levels:

Pain study: $n = 16$, $\bar{x} = 45.2$, $s = 8.4$, $\text{SE} = 2.10$

Confidence	α	$t_{\alpha/2,15}$	ME	CI
90%	0.10	1.753	3.68	(41.5, 48.9)
95%	0.05	2.131	4.48	(40.7, 49.7)
99%	0.01	2.947	6.19	(39.0, 51.4)

The confidence-precision trade-off:

- More confident $\Rightarrow$ wider interval (less precise)
- Less confident $\Rightarrow$ narrower interval (more precise)
- 95% is conventional in most fields
- 99% for high-stakes decisions (e.g., safety-critical applications)
- 90% for preliminary or exploratory studies

Effect of Variability

Same sample size and confidence, different variability:

All samples: $n = 25$, $\bar{x} = 50$, 95% confidence, $t_{0.975,24} = 2.064$

Scenario	s	SE	ME	95% CI
Low variability	5	1.00	2.06	(48.0, 52.0)
Moderate variability	10	2.00	4.13	(45.9, 54.1)
High variability	20	4.00	8.26	(41.7, 58.3)

Key insight:

- Variability directly impacts precision
- Cannot change s through study design (it's a population property)
- Can only counteract high s by increasing n
- More homogeneous populations give more precise estimates

CI for a Population Proportion

When the outcome is binary:

- Success/failure, yes/no, disease/no disease
- Estimate population proportion p
- Sample proportion: $\hat{p} = \frac{x}{n}$ (where x = number of successes)

Standard error for proportion:

$$\text{SE} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

95% Confidence interval (Wald method):

$$\hat{p} \pm z_{0.975} \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\hat{p} \pm 1.96 \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Note: Uses Z (not t) because with large n, proportion is approximately

When to Use Normal Approximation for Proportions

The normal approximation works well when:

$$n\hat{p} \geq 10 \quad \text{and} \quad n(1-\hat{p}) \geq 10$$

This ensures:

- At least 10 successes
- At least 10 failures
- Sampling distribution of $\hat{p}$ is approximately normal

If conditions not met:

- Use exact binomial methods (beyond this course)
- Or use Wilson score interval (more advanced)
- Or collect more data

Common in biostatistics:

- Disease prevalence
- Treatment success rates

Example: Vaccine Efficacy

Clinical trial: New vaccine tested on 200 people

- 174 developed immunity (positive antibody response)
- Sample proportion: $\hat{p} = \frac{174}{200} = 0.87$

Check conditions:

- $n\hat{p} = 200 \times 0.87 = 174 \geq 10$ ✓
- $n(1 - \hat{p}) = 200 \times 0.13 = 26 \geq 10$ ✓

Calculate 95% CI:

Standard error: $SE = \sqrt{\frac{0.87 \times 0.13}{200}} = \sqrt{\frac{0.1131}{200}} = \sqrt{0.0005655} = 0.0238$

Margin of error: $ME = 1.96 \times 0.0238 = 0.047$

95% CI: $0.87 \pm 0.047 = (0.823, 0.917)$ **or (82.3%, 91.7%)**

Interpretation: We are 95% confident that the true vaccine efficacy is between 82.3% and 91.7%.

Example: Disease Prevalence

Public health survey: Screen 500 adults for diabetes

- 65 test positive
- Sample proportion: $\hat{p} = \frac{65}{500} = 0.13$

Check conditions:

- $n\hat{p} = 500 \times 0.13 = 65 \geq 10$ ✓
- $n(1 - \hat{p}) = 500 \times 0.87 = 435 \geq 10$ ✓

Calculate 95% CI:
SE: $\sqrt{\frac{0.13 \times 0.87}{500}} = \sqrt{0.0002262} = 0.0150$
ME: $1.96 \times 0.0150 = 0.0294$
95% CI: $0.13 \pm 0.029 = (0.101, 0.159)$ **or (10.1%, 15.9%)**

Interpretation: We estimate that between 10.1% and 15.9% of adults in this population have diabetes.

Planning Studies: How Large a Sample?

Before collecting data, researchers ask:

- How many subjects do we need?
- What precision do we want?
- How wide will our confidence interval be?

Sample size determination:

- Work backwards from desired margin of error
- Requires estimate of population variability
- Balances precision with cost/feasibility

General approach:

1. Specify desired margin of error (ME)
2. Estimate population SD (σ) from pilot data or literature
3. Choose confidence level (usually 95%)
4. Solve for n in the ME equation

Starting with margin of error formula:

$$\text{ME} = z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

Solve for n:

$$\sqrt{n} = \frac{z_{\alpha/2} \times \sigma}{\text{ME}}$$

$$n = \left(\frac{z_{\alpha/2} \times \sigma}{\text{ME}}\right)^2$$

For 95% confidence:

$$n = \left(\frac{1.96 \times \sigma}{\text{ME}}\right)^2$$

Note:

- Uses Z (not t) for planning
- Round up to next integer

Example: Cholesterol Study Planning

Goal: Estimate mean cholesterol with margin of error $= 5$ mg/dL

Given information:

- Desired ME $= 5$ mg/dL
- From literature: $\sigma \approx 40$ mg/dL
- Confidence level: 95% (so $z_{0.975} = 1.96$)

Calculate required sample size:

$$n = \left(\frac{1.96 \times 40}{5}\right)^2 = \left(\frac{78.4}{5}\right)^2 = (15.68)^2 = 245.86$$

Round up: $n = 246$ subjects

Interpretation:

- Need to sample 246 people
- Will give 95% CI with margin of error ± 5 mg/dL
- Actual CI might be slightly different (using s instead of σ)

Example: Blood Pressure Study Planning

Different margins of error:

Suppose $\sigma = 15$ mmHg for blood pressure, 95% confidence

Desired ME	Calculation	Required n
2 mmHg	$(1.96 \times 15/2)^2$	217
3 mmHg	$(1.96 \times 15/3)^2$	97
5 mmHg	$(1.96 \times 15/5)^2$	35
10 mmHg	$(1.96 \times 15/10)^2$	9

Key insights:

- Halving ME requires $4\times$ the sample size
- More precision is expensive (in terms of sample size)
- Diminishing returns: going from ME=5 to ME=2 adds 182 subjects
- Practical considerations: balance precision with feasibility

Sample Size for Estimating a Proportion

For proportions:

$$n = \left(\frac{z_{\alpha/2}}{\text{ME}}\right)^2 p(1-p)$$

Problem: We need p to calculate n, but we don't know p yet!

Three approaches:

1. **Use pilot data or prior estimate:**
   - Previous studies suggest $p \approx 0.3$
   - Use this value in the formula
2. **Conservative approach:**
   - Use $p = 0.5$ (maximizes $p(1-p)$)
   - Gives largest possible n (safe but possibly wasteful)
3. **Range of plausible values:**
   - Calculate n for several plausible p values
   - Choose based on context

Example: Prevalence Study Planning

Goal: Estimate disease prevalence with ME $= 0.03$ (3%), 95% confidence

Scenario 1: No prior information (use $p = 0.5$)

$$n = \left(\frac{1.96}{0.03}\right)^2 \times 0.5 \times 0.5 = (65.33)^2 \times 0.25 = 1067.1$$

Need $n = 1068$ subjects

Scenario 2: Prior studies suggest $p \approx 0.15$

$$n = \left(\frac{1.96}{0.03}\right)^2 \times 0.15 \times 0.85 = (65.33)^2 \times 0.1275 = 544.5$$

Need $n = 545$ subjects

Comparison:

- Conservative approach requires nearly $2\times$ more subjects
- If we have good prior information, can save resources
- Trade-off: certainty in planning vs. sample size efficiency

Large Sample vs Small Sample CIs

When n is large ($n \geq 30$ as rule of thumb):

- t-distribution $\approx$ normal distribution
- Can use $z \approx 1.96$ instead of t for 95% CI
- CLT ensures $\bar{X}$ is approximately normal
- Less sensitive to departures from normality

When n is small ($n < 30$):

- Must use t-distribution (heavier tails)
- Need stronger normality assumption
- Check data for severe skewness or outliers
- Consider transformations if badly non-normal

Practical guideline:

- Always use t-distribution when σ unknown
- Check normality assumption with histogram or Q-Q plot
- For very small n (< 15), be especially careful
- Robust methods exist for non-normal data (beyond this course)

Using CIs for Hypothesis Testing

CIs can answer yes/no questions:

Example: Is the population mean different from a specific value?

Drug efficacy study:

- Research question: Does the drug reduce pain by more than 30 points on average?
- 95% CI for mean reduction: (35.2, 48.6)
- Since entire CI is above 30, YES, strong evidence

General rule for 95% CI:

- If hypothesized value is *outside* the CI: reject at $\alpha = 0.05$
- If hypothesized value is *inside* the CI: fail to reject at $\alpha = 0.05$
- CI gives more information than p-value (shows magnitude and precision)

Next week: Formal hypothesis testing procedures

Common Mistakes with Confidence Intervals

1. **Misinterpreting the confidence level**
   - $\times$ "95% probability that μ is in this interval"
   - ✓ "95% of such intervals contain μ"
2. **Confusing CI for mean with range of data**
   - CI estimates where population mean is
   - Not where individual observations fall
3. **Ignoring assumptions**
   - Random sampling required
   - Independence required
   - Approximate normality (or large n)
4. **Using Z when should use t**
   - Always use t when σ is unknown (almost always!)
5. **Forgetting to round n up in planning**
   - $n = 245.86$ means need 246 subjects, not 245

Checking Assumptions: Normality

How to assess if data are approximately normal:

1. **Histogram**
   - Should be roughly bell-shaped
   - Check for severe skewness or multiple modes
2. **Q-Q plot (quantile-quantile plot)**
   - Points should fall roughly on a straight line
   - Deviations at ends less concerning than in middle
3. **Formal tests (Shapiro-Wilk)**
   - Tests null hypothesis that data are normal
   - Very sensitive with large n (may reject for trivial departures)

Remember:

- t-procedures are robust to mild departures
- Larger n means less sensitive to non-normality (CLT)
- Outliers more problematic than skewness

AI for Confidence Interval Analysis

How AI can help this week:

- **CI calculations:** Computing intervals step-by-step
- **Sample size planning:** Determining required n
- **Interpretation:** Writing correct interpretations
- **Assumption checking:** Creating diagnostic plots
- **Comparison:** Different confidence levels or methods
- **R code:** Using `t.test()` and related functions

Example prompts:

- "Calculate 95% CI for mean given: n=20, mean=105, SD=15"
- "How many subjects needed for ME=3 if SD=12 at 95% confidence?"
- "Create R code to check normality assumption with histogram and Q-Q plot"
- "Explain the difference between 90%, 95%, and 99% confidence intervals"

AI Demo: CI Calculation in R

Prompt: "Calculate 95% CI for cholesterol data in R"

Claude AI Response:

```
# Cholesterol data (mg/dL)
cholesterol <- c(205, 198, 220, 185, 210, 195, 225, 200, 215, 192,
                 208, 196, 218, 203, 212, 189, 207, 201, 213, 199)

# Calculate sample statistics
n <- length(cholesterol)
xbar <- mean(cholesterol)
s <- sd(cholesterol)
se <- s / sqrt(n)

# Calculate 95% CI using t-distribution
df <- n - 1
t_crit <- qt(0.975, df)
me <- t_crit * se
ci_lower <- xbar - me
ci_upper <- xbar + me

# Display results
cat("Sample size:", n, "\n")
cat("Sample mean:", round(xbar, 2), "mg/dL\n")
cat("Sample SD:", round(s, 2), "mg/dL\n")
cat("Standard error:", round(se, 3), "mg/dL\n")
cat("t critical value:", round(t_crit, 3), "\n")
cat("Margin of error:", round(me, 2), "mg/dL\n")
cat("95% CI: (", round(ci_lower, 2), ", ", round(ci_upper, 2), ")\n", sep="")

# Alternative: use t.test() function
t.test(cholesterol, conf.level=0.95)
```

AI Demo: Sample Size Calculation

Prompt: "Calculate sample size needed for blood pressure study in R"

Claude AI Response:

```
# Sample size calculation for estimating mean BP
# Goal: 95% CI with margin of error = 3 mmHg
# Estimated population SD = 15 mmHg

sigma_est <- 15     # Estimated population SD
me_desired <- 3     # Desired margin of error
conf_level <- 0.95  # Confidence level
alpha <- 1 - conf_level
z_crit <- qnorm(1 - alpha/2)  # 1.96 for 95%

# Calculate required sample size
n_required <- (z_crit * sigma_est / me_desired)^2

cat("Sample size calculation:\n")
cat("Estimated SD:", sigma_est, "mmHg\n")
cat("Desired margin of error:", me_desired, "mmHg\n")
cat("Confidence level:", conf_level*100, "%\n")
cat("Z critical value:", round(z_crit, 3), "\n\n")
cat("Required sample size (exact):", round(n_required, 2), "\n")
cat("Required sample size (rounded up):", ceiling(n_required), "\n")

# Verify the calculation
actual_me <- z_crit * sigma_est / sqrt(ceiling(n_required))
cat("\nVerification - actual ME with n =", ceiling(n_required), ":",
    round(actual_me, 3), "mmHg\n")
```

AI Demo: Comparing Confidence Levels

Prompt: “Compare 90%, 95%, 99% CIs for same data in R”

Claude AI Response:

```
# Sample data: pain reduction scores
pain_reduction <- c(42, 38, 51, 45, 39, 48, 44, 50, 43, 47,
                    41, 46, 49, 40, 52)

# Calculate sample statistics
n <- length(pain_reduction)
xbar <- mean(pain_reduction)
s <- sd(pain_reduction)
se <- s / sqrt(n)
df <- n - 1

# Calculate CIs for different confidence levels
conf_levels <- c(0.90, 0.95, 0.99)

cat("Sample mean:", round(xbar, 2), "\n")
cat("Sample SD:", round(s, 3), "\n")
cat("Sample size:", n, "\n")
cat("Standard error:", round(se, 3), "\n\n")

for (conf in conf_levels) {
  alpha <- 1 - conf
  t_crit <- qt(1 - alpha/2, df)
  me <- t_crit * se
  ci_lower <- xbar - me
  ci_upper <- xbar + me
  width <- ci_upper - ci_lower

  cat(conf*100, "% CI: (", round(ci_lower, 2), ", ",
```

AI Demo: Checking Normality Assumption

Prompt: "Create diagnostic plots to check normality assumption"

Claude AI Response:

```
# Sample data
data <- c(23, 25, 28, 22, 27, 24, 26, 29, 25, 23, 28, 24, 27, 25, 26)

# Create diagnostic plots
par(mfrow=c(1,2))  # 1 row, 2 columns

# 1. Histogram with normal curve overlay
hist(data, breaks=8, prob=TRUE, col="lightblue",
     main="Histogram with Normal Curve",
     xlab="Value", border="white")
# Overlay theoretical normal curve
xfit <- seq(min(data), max(data), length=100)
yfit <- dnorm(xfit, mean=mean(data), sd=sd(data))
lines(xfit, yfit, col="red", lwd=2)

# 2. Q-Q plot
qqnorm(data, main="Q-Q Plot", pch=19, col="blue")
qqline(data, col="red", lwd=2)

par(mfrow=c(1,1))  # Reset layout

# Formal normality test
shapiro_test <- shapiro.test(data)
cat("\nShapiro-Wilk test for normality:\n")
cat("W statistic:", round(shapiro_test$statistic, 4), "\n")
cat("p-value:", round(shapiro_test$p.value, 4), "\n")
if (shapiro_test$p.value > 0.05) {
  cat("Conclusion: No evidence against normality (p > 0.05)\n")
```

AI Best Practices for CIs

DO:

- Verify calculations manually first
- Check assumptions before computing CI
- Ask for interpretations
- Request visualizations
- Compare different methods
- Use AI to learn concepts

DON'T:

- Trust results blindly
- Skip assumption checking
- Confuse Z and t methods
- Ignore sample size requirements
- Forget to interpret in context
- Use without understanding

Key Verification Steps:

1. Check that sample statistics are correct
2. Verify correct critical value (t not Z, correct df)
3. Confirm SE calculation
4. Ensure interpretation makes sense in context
5. Cross-check with `t.test()` output

R Example: Basic t-Interval

```
# Blood pressure data (mmHg)
bp <- c(128, 132, 125, 138, 130, 135, 127, 140,
        129, 133, 136, 126, 131, 134, 137)

# Method 1: Manual calculation
xbar <- mean(bp)
s <- sd(bp)
n <- length(bp)
se <- s / sqrt(n)
df <- n - 1
t_crit <- qt(0.975, df)  # 95% CI
me <- t_crit * se
ci <- c(xbar - me, xbar + me)

cat("Manual: 95% CI =", round(ci, 2), "\n")

# Method 2: Using t.test()
result <- t.test(bp, conf.level=0.95)
cat("t.test(): 95% CI =", round(result$conf.int, 2), "\n")
```

R Example: CI for Proportion

```
# Vaccine study: 174 out of 200 developed immunity
x <- 174         # Number of successes
n <- 200         # Sample size
p_hat <- x / n   # Sample proportion

# Check conditions
cat("np =", n*p_hat, ", n(1-p) =", n*(1-p_hat), "\n")

# Calculate 95% CI
se <- sqrt(p_hat * (1 - p_hat) / n)
z_crit <- qnorm(0.975)  # 1.96
me <- z_crit * se
ci_lower <- p_hat - me
ci_upper <- p_hat + me

cat("Sample proportion:", p_hat, "\n")
cat("Standard error:", round(se, 4), "\n")
cat("95% CI: (", round(ci_lower, 4), ", ",
    round(ci_upper, 4), ")\n", sep="")
```

R Example: Sample Size Planning

```r
# Function to calculate required sample size for mean
sample_size_mean <- function(sigma, me, conf.level=0.95) {
  alpha <- 1 - conf.level
  z <- qnorm(1 - alpha/2)
  n <- (z * sigma / me)^2
  return(ceiling(n))  # Round up
}

# Example: Cholesterol study
sigma_est <- 40   # mg/dL
me_desired <- 5   # mg/dL

n_needed <- sample_size_mean(sigma_est, me_desired, 0.95)
cat("Required sample size:", n_needed, "\n")

# Try different margins of error
margins <- c(2, 3, 5, 10)
for (me in margins) {
  n <- sample_size_mean(sigma_est, me, 0.95)
  cat("ME =", me, "-> n =", n, "\n")
}
```

R Example: Visualizing Multiple CIs

```
# Simulate multiple samples and CIs
set.seed(524)
true_mean <- 100   # True population mean
true_sd <- 15      # True population SD
n <- 25            # Sample size
num_samples <- 20  # Number of samples to draw

# Storage for CIs
ci_data <- data.frame(sample=1:num_samples,
                      lower=NA, upper=NA, contains_mu=NA)

for (i in 1:num_samples) {
  sample_data <- rnorm(n, true_mean, true_sd)
  result <- t.test(sample_data, conf.level=0.95)
  ci_data$lower[i] <- result$conf.int[1]
  ci_data$upper[i] <- result$conf.int[2]
  ci_data$contains_mu[i] <- (ci_data$lower[i] <= true_mean) &
                            (true_mean <= ci_data$upper[i])
}

# Plot the CIs
```

Chapter 9 Key Concepts:

- **Confidence intervals:** Range of plausible values for parameter
  - General form: Estimate $\pm$ (Critical value) $\times$ SE
- **t-interval for mean:** $\bar{x} \pm t_{\alpha/2,n-1} \times \frac{s}{\sqrt{n}}$
  - Use when σ unknown (almost always)
  - Requires approximate normality or large n
- **CI for proportion:** $\hat{p} \pm z_{\alpha/2} \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
  - Requires $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$
- **Factors affecting width:**
  - Sample size n (larger $\rightarrow$ narrower)
  - Variability s (larger $\rightarrow$ wider)
  - Confidence level (higher $\rightarrow$ wider)
- **Sample size planning:** Work backwards from desired ME

Next Week: Hypothesis testing (one-sample and two-sample tests)

Questions?

Thank you!

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